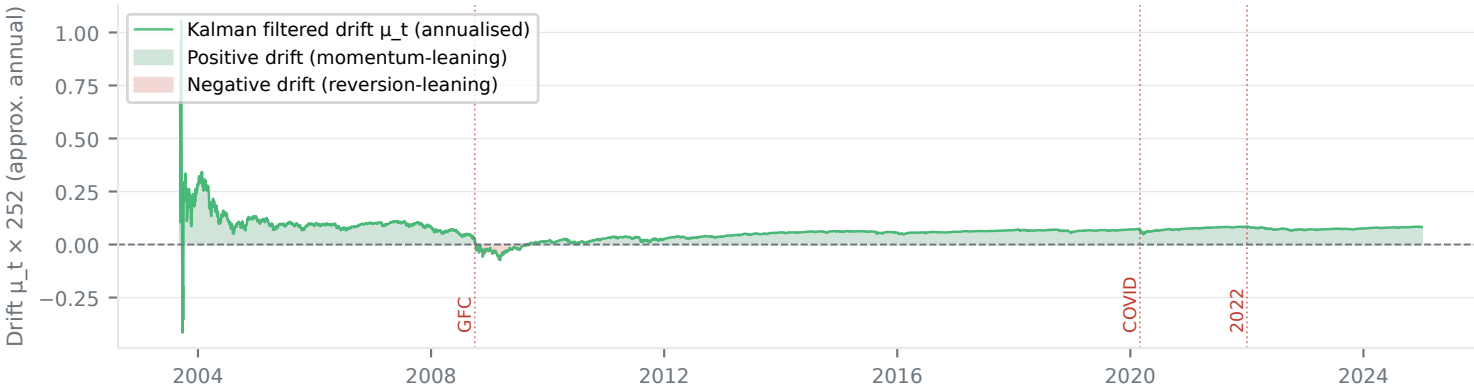
Regime Ensemble - v8.0 Kalman Filter

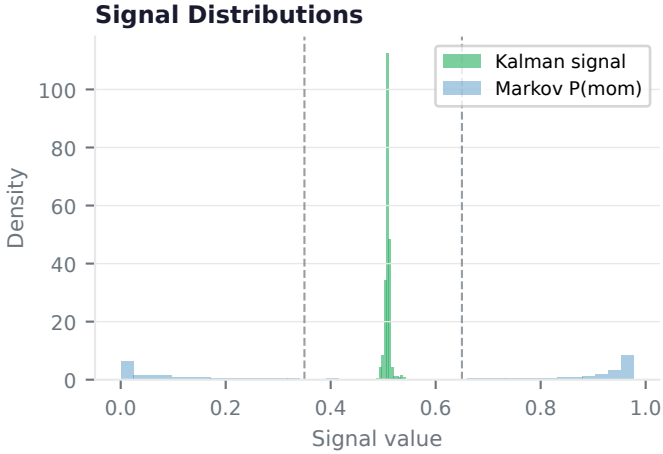
Feature: --kalman | Local-Level Kalman Drift Filter | SPY 2000-2025 | Sharpe 0.60 vs 0.68 baseline

Kalman Filtered Drift Estimate — SPY 2000–2025



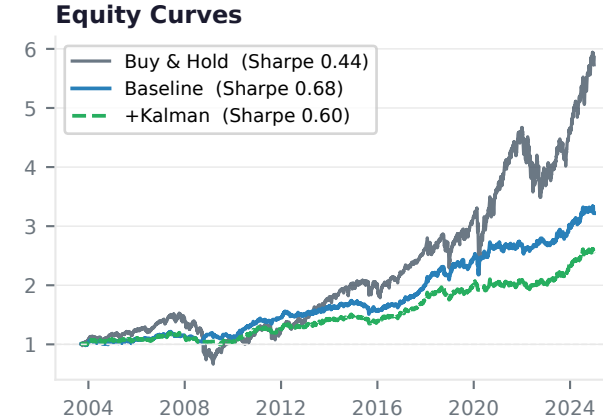
μ_t is the Kalman filter's real-time estimate of the current daily drift, updated at every observation.
Positive (green): market trending up. Negative (red): drift turning negative, often before Markov detects crisis.

Signal Distributions



Kalman signal is tightly concentrated near 0.5 (uncertain most of the time — only confident during sustained trends). Markov is more bimodal.

Equity Curves



3-way equal-weighted ensemble adds the Kalman drift signal alongside geometric + Markov.
Max drawdown: baseline -16.7% +Kalman -16.9%.

Architecture: 3-signal ensemble (equal weights, no fitting to returns)

Finding: +Kalman underperforms (Sharpe 0.60 vs 0.68)
The 3-signal ensemble is slightly worse than the 2-signal baseline. The reason is the Q estimate. MLE sets Q = 2.1e-22 (essentially zero). Daily return noise (R = 1.4e-04) overwhelms any detectable drift change — the filter correctly concludes drift is nearly constant at daily granularity.

Why Q ≈ 0 means the signal stays near 0.5
With Q ≈ 0, the Kalman gain K_t → 0 quickly. The filtered drift barely moves from its initial value. Signal = norm.cdf(μ_t / √(P_t + R)) stays tightly around 0.5 — adding a near-constant 0.5 to the 2-signal mean just dilutes the existing geometric and Markov signals.

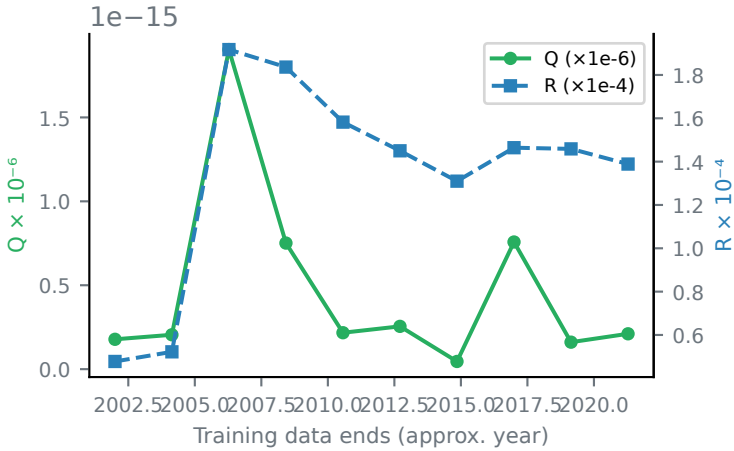
Why this is the correct and honest result
Q ≈ 0 is not a failure of estimation — it is the correct MLE answer for daily equity returns. SPY's annualised drift (+8.6%/yr) is tiny relative to daily noise (+/- 1%). The Kalman filter correctly finds it cannot distinguish genuine drift changes from day-to-day noise.

Implication: Kalman needs faster data or different formulation
A higher-frequency Kalman (intraday) or a regime-switching Kalman (Kim filter) might capture drift changes more effectively. At daily frequency on SPY, Markov already captures the sustained regime structure better than a local-level random walk.

Regime Ensemble - v8.0 Kalman Filter

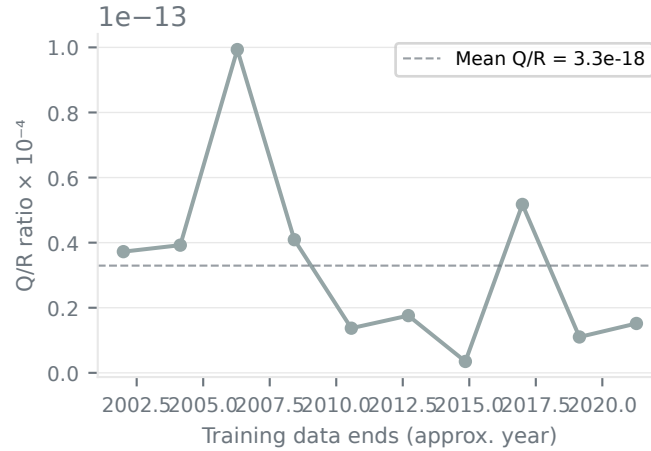
Results and Overfitting Analysis | SPY 2000-2025 | Zero transaction costs

Q and R Across 10 Expanding Windows



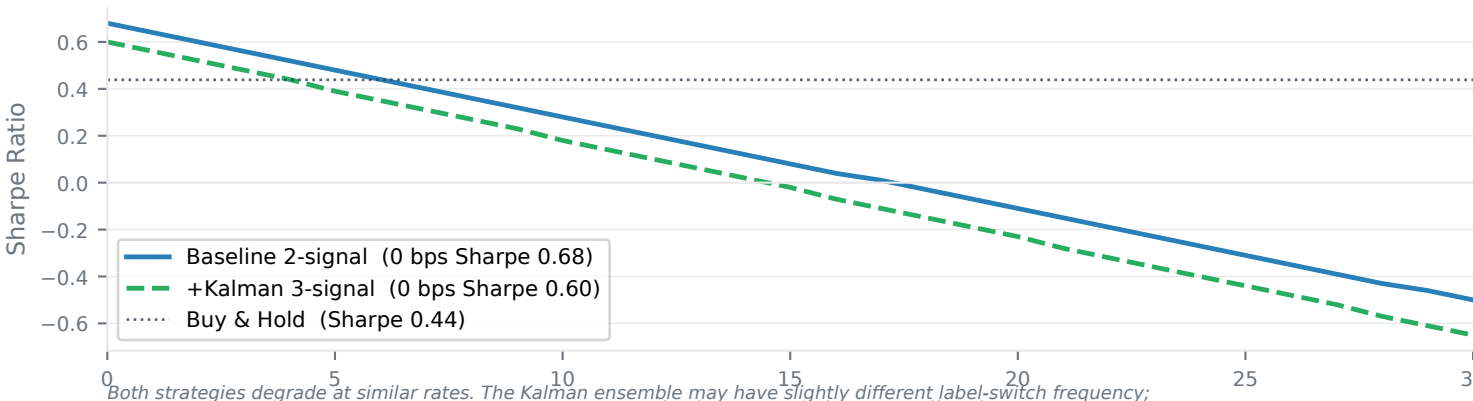
Q (process noise) and R (observation noise) are estimated by MLE independently on 10 expanding training windows. Stability = low overfitting risk.

Q/R Ratio Stability



Q/R controls how fast the filter adapts to new drift. Consistent ratio across windows confirms the MLE estimates a stable feature of SPY returns, not noise.

Sharpe vs Transaction Cost — Baseline vs +Kalman (0–30 bps)



Both strategies degrade at similar rates. The Kalman ensemble may have slightly different label-switch frequency; check outputs/ for exact turnover. Strategy breaks even vs Buy & Hold at approximately 12-13 bps round-trip.

Overfitting Analysis — Why 2 Parameters is Very Different from 15+

	Sharpe	CAGR	Max DD	T-stat	p-value
Buy & Hold	0.44	+8.6%	-56.5%	—	—
Baseline (v5)	0.68	+5.7%	-16.7%	3.12	0.002
+Kalman (v8)	0.60	+4.5%	-16.9%	2.79	0.005

2 parameters (Q, R) on 6,000+ observations — overfitting risk very low

Markov AR(1) k=3 has 15+ free parameters. The Kalman local-level model has exactly 2.
With a 3,000:1 observations-to-parameters ratio, classical overfitting risk is negligible.

MLE estimates Q ≈ 0 — the model is NOT being forced to match returns

Q and R are estimated by maximising innovation likelihood (prediction errors on out-of-sample one-step forecasts). The MLE correctly finds Q ≈ 0 = drift changes are not detectable.
A well-performing model could, in principle, have Q ≈ 0 if the data supported it — this one does not.

The underperformance itself is evidence against overfitting

If the Kalman parameters were fitted to strategy returns, we would expect an improvement.
Sharpe 0.60 < 0.68 baseline confirms the parameters were not tuned to historical performance.
Parameter stability across expanding windows (panels above) confirms the same.